

CHAPTER 1

INTRODUCTION

1.1 OVERVIEW

The College Admission Enquiry System is a cloud-based application developed using the Salesforce Customer Relationship Management (CRM) platform to simplify and manage the college admission process digitally. The main objective of this project is to replace traditional manual admission procedures with an automated online system. In many colleges, the admission process involves paperwork, manual verification, and multiple administrative steps, which can lead to delays, errors, and difficulty in tracking applications. This project provides a centralized platform where students can easily interact with the admission system and complete the required procedures online.

The system allows students to check seat availability, verify eligibility, submit admission enquiries, upload required documents, pay admission fees, and receive seat confirmation through an online portal. The application is developed using Lightning Web Components (LWC) to design the user interface and Apex programming to handle server-side business logic and database operations. Students access the system through a Salesforce Experience Cloud portal, which provides a user-friendly and responsive interface that can be accessed from computers or mobile devices.

The system includes several components such as seat availability checking, eligibility verification, enquiry form submission, application tracking, and fee payment. It also provides real-time status updates to students, reducing the need for physical visits and manual follow-ups. The system uses Salesforce automation tools such as Record-Triggered Flows to automatically send document request notifications, confirm admissions, and update the available seat count when a student secures admission.

1.2 EXISTING SYSTEM

The existing college admission system mainly follows a manual and paper-based process, which creates several difficulties for both students and administrative staff. Students must visit the college campus to collect admission brochures and application forms, making the process inconvenient for those living far from the institution. They are required to fill out forms manually and submit photocopies of academic documents such as 10th and 12th mark sheets, transfer certificates, and conduct certificates directly at the administration office. This physical submission process takes time and increases the chances of delays and document misplacement.

After receiving the applications, admission staff check student eligibility by manually comparing academic marks with the required criteria. The results are recorded in registers or files, which makes the process slow and prone to human errors. Students must then visit the college again for document verification, where original certificates are checked against submitted copies. Managing large numbers of paper files makes storage and retrieval difficult and increases the risk of losing important records.

The fee payment process is handled manually through bank challans, demand drafts, or cash payments at the college accounts office. Payment details are recorded in registers, making it difficult to link payments accurately with specific applications. Communication between students and the college is limited, as updates about admission status are usually shared through notice boards, phone calls, or postal messages. Students often need to contact the office repeatedly to know the progress of their application.

Seat availability is tracked using manual spreadsheets maintained by the administration staff. Since updates are not automatic, there is a risk of incorrect seat allocation during peak admission periods. Overall, the manual admission system results in slow processing, increased workload for staff, and inconvenience for students.

1.3 DRAWBACKS OF EXISTING SYSTEM

The existing manual admission system exhibits the following critical drawbacks that collectively justify the development of the proposed digital solution:

- Students are required to visit the institution multiple times for brochure collection, form submission, document verification, fee payment, and admission confirmation, which makes the admission process time-consuming.
- The requirement of physical presence creates accessibility problems for students from remote locations or those with mobility and financial constraints.
- Manual verification of student eligibility by administration staff increases the chances of human errors and inconsistent admission decisions.
- Students cannot check the real-time status of their application and must rely on phone calls or campus visits to get updates.
- The manual admission process cannot efficiently handle a large number of applications during peak admission periods.
- Physical storage of student documents increases the risk of loss, damage, or misplacement of important records.
- Communication between the institution and students is slow because updates are shared through notice boards, phone calls, or postal messages.
- The system does not provide automated reports or analytical insights for administrators to monitor admission progress.
- Seat availability is tracked manually, which may lead to incorrect seat allocation when multiple applications are processed simultaneously.
- Administration staff spend a large amount of time performing repetitive manual tasks such as data entry, document checking, and handling enquiries.

These drawbacks result in delays in the admission process, increased administration workload, higher chances of errors, and reduced convenience for students applying for admission.

1.4 PROPOSED SYSTEM

The proposed College Admission Enquiry System is a cloud-based digital solution developed to replace the manual admission process with an automated system using the Salesforce platform. It integrates all stages of admission into a single platform where students and administrators can interact through role-based access. This centralized system improves efficiency, reduces manual work, and ensures that admission activities are handled in a more organized and transparent manner.

Students access the system through an online portal built using Salesforce Experience Cloud, which can be used from any device with an internet connection. The portal allows students to view seat availability in different departments and check their eligibility based on academic performance. It provides a simple and user-friendly interface that helps students easily understand the admission process and make informed decisions without visiting the institution.

The system provides an online enquiry form where students can submit their personal and academic details. After submission, the system automatically evaluates eligibility and generates a unique enquiry number for tracking. Students can log in to track their application status, submit required documents, and complete the fee payment process. Automated notifications are sent to students at each stage, ensuring they stay updated about their application progress.

Once the fee is verified, the system confirms the admission and automatically updates the seat availability for the selected department. The system maintains accurate records of all admission activities, ensures secure data storage, and reduces the chances of errors. In addition, reports and dashboards help administrators monitor admission progress, analyze data, and manage the overall admission process effectively, making the system reliable, scalable, and easy to maintain. The platform supports easy data retrieval and quick decision-making through real-time insights. It enhances transparency by allowing students to track every stage of their application. Overall, the system provides a modern and efficient solution for managing the entire admission process.

1.5 ADVANTAGES OF PROPOSED SYSTEM

The proposed system delivers the following significant advantages over the existing manual process:

- The entire admission process is completed digitally, removing the need for paper forms, physical registers, and manual documentation.
- Students can access the admission portal at any time from any location using an internet-connected device.
- Eligibility of students is evaluated automatically using formula-based calculations, ensuring consistent and error-free assessment.
- Seat availability is updated automatically when an admission is confirmed, ensuring accurate and real-time seat count.
- Automated email notifications are sent to students at important stages such as document submission and admission confirmation.
- All submitted documents are stored in a structured digital system, making them easy to manage, search, and verify.
- The system provides secure access through role-based permissions, ensuring that users can only access information relevant to their role.
- The cloud-based infrastructure allows the system to handle a large number of applications without affecting performance.
- Administrators can monitor admission data through real-time reports and dashboards for better decision-making.
- The system maintains a complete record of all activities and changes, helping with accountability and future reference.

The proposed College Admission Enquiry System improves efficiency by automating the entire admission process and reducing manual work. It provides a faster, more reliable, and accessible platform for both students and administrators to manage admissions effectively.

SUMMARY

In this chapter, the existing college admission process and its limitations were examined, highlighting challenges such as time-consuming manual procedures, lack of real-time application tracking, manual eligibility verification, and inefficient communication between students and administrators. The drawbacks of the traditional paper-based admission system were analyzed, emphasizing the need for a centralized digital solution to improve efficiency, accessibility, and accuracy in managing admission activities.

The proposed College Admission Enquiry System was introduced as a cloud-based solution designed to automate and streamline the entire admission process. The system enables students to check seat availability, verify eligibility, submit enquiries, upload documents, and complete fee payments through an online platform. By utilizing the Salesforce platform with features such as automation, real-time data management, and secure access control, the system improves the overall admission experience for both students and administrators. Chapter 2 presents the system requirements and explains the technologies used for developing and implementing the proposed solution.

CHAPTER 2

SYSTEM ANALYSIS

2.1 IDENTIFICATION OF NEED

The need for the College Admission Enquiry System was identified after analyzing the challenges faced in the traditional manual admission process. Observations of admission procedures, discussions with administration staff and students, and the review of previous admission records revealed several operational issues such as delays, errors in eligibility verification, difficulty in tracking applications, and heavy administration workload.

One of the major needs identified was the removal of physical barriers in the admission process. Many students from remote areas faced difficulties in visiting the institution multiple times for application submission, document verification, and fee payment. Therefore, a fully digital system accessible from any location was required to make the admission process more convenient and inclusive.

Another important need was the automation of eligibility verification and communication with applicants. Manual evaluation of student marks often resulted in inconsistencies and increased processing time. An automated system that can instantly evaluate eligibility, send notifications, and allow students to track their application status was necessary to improve efficiency and reduce staff workload.

The analysis also highlighted the need for better document management, secure data storage, and real-time reporting. A digital platform can store student documents safely, provide accurate admission statistics to administrators, and support future scalability as the number of applicants increases. Therefore, implementing a cloud-based admission enquiry system was identified as an effective solution to address these challenges.

2.2 FEASIBILITY STUDY

A feasibility study is an important stage in system analysis because it helps determine whether the proposed system can be successfully developed and implemented within the available resources and constraints. It evaluates the practicality, effectiveness, and sustainability of the system before the actual development process begins. Conducting a feasibility study ensures that the proposed project is realistic, beneficial, and capable of meeting the needs of its users.

In this project, the feasibility study is conducted to analyze the proposed College Admission Enquiry System from three main perspectives: economic feasibility, operational feasibility, and technical feasibility. Economic feasibility examines whether the benefits provided by the system justify the costs involved in development, implementation, and maintenance. Operational feasibility focuses on how effectively the system can function within the existing admission process of the institution and whether students and administration staff can easily use and adapt to the new system. Technical feasibility evaluates whether the required technology, software platform, and technical expertise are available to develop and maintain the system successfully.

Three considerations involved in feasibility are:

- Economic Feasibility
- Operational Feasibility
- Technical Feasibility

2.2.1 Economic Feasibility

Economic feasibility evaluates whether the proposed system is cost-effective and beneficial compared to its development and maintenance costs. The College Admission Enquiry System is economically feasible as it is built on the Salesforce cloud platform, eliminating the need for additional hardware and reducing infrastructure costs. It minimizes expenses related to paperwork and manual processing while improving efficiency and saving time, making it a cost-effective solution.

2.2.2 Operational Feasibility

Operational feasibility refers to the evaluation of how effectively the proposed system can be implemented and used within the existing operational environment. It examines whether the system will function properly in real-world conditions and whether the users of the system can easily adapt to it.

The proposed college admission enquiry System is operationally feasible because it is designed to be simple, user-friendly, and accessible to both students and administration staff. Students can easily access the online portal to check seat availability, verify eligibility, submit enquiries, upload documents, and track their application status. The clear interface and structured admission process make it easy for users to navigate the system without requiring advanced technical knowledge.

2.2.3 Technical Feasibility

Technical feasibility evaluates whether the required technical resources, tools, and expertise are available to develop and maintain the proposed system. It ensures that the selected technologies are reliable, stable, and capable of meeting the system's functional requirements.

The proposed College Admission Enquiry System is technically feasible because it is developed using the Salesforce platform, which is a well-established and widely used cloud-based application development environment. The system uses Lightning Web Components for building the user interface, Apex for implementing server-side logic, and Salesforce automation tools such as Flows for handling automated processes. These technologies are well-documented, actively supported, and widely used in enterprise application development.

2.3 SOFTWARE REQUIREMENT SPECIFICATIONS

Software Requirement Specification (SRS) defines the functional and non-functional requirements of the proposed system. It provides a clear description of how the system should operate and outlines the software and hardware resources required for successful implementation. The SRS acts as a guideline for system design, development, testing, and deployment, ensuring that the final system meets the expected objectives and user requirements.

2.3.1 Software Requirements

The following software requirements are necessary for the development and execution of the proposed system:

- **Operating System:** Windows 10 / Windows 11 / macOS / Linux
- **Development Environment:** Visual Studio Code with Salesforce Extension Pack
- **Salesforce Platform:** Salesforce Developer Edition (API Version 59.0 or later)
- **Frontend Technology:** Lightning Web Components (LWC)
- **Backend Technology:** Salesforce Apex
- **Query Language:** SOQL (Salesforce Object Query Language)
- **Portal Platform:** Salesforce Experience Cloud
- **Automation Tool:** Salesforce Record-Triggered Flows
- **Email Notification System:** Salesforce Email Alerts with HTML Email Templates
- **Web Browser:** Google Chrome, Mozilla Firefox, Microsoft Edge
- **Development Tools:** Salesforce CLI (SFDX), Visual Studio Code

These software tools and technologies support efficient development, testing, deployment, and maintenance of the College Admission Enquiry System.

2.3.2 Hardware Requirements

The following hardware requirements are necessary for the development and execution of the proposed system. Since the College Admission Enquiry System is developed on the Salesforce cloud platform, no dedicated server hardware or database infrastructure is required by the institution. The system can be accessed through standard computing devices with an internet connection.

- **Device Type:** Laptop / Desktop Computer / Tablet / Mobile Device
- **Processor:** Intel Core i5 / AMD Ryzen 5 or higher Processor Speed: 2.0 GHz or higher
- **Installed RAM:** Minimum 8 GB (4 GB minimum required)
- **System Type:** 64-bit Operating System, x64-based Processor
- **Storage:** Minimum 256 GB Hard Disk / SSD

- **Display Resolution:** 1366 × 768 pixels or higher
- **Internet Connection:** Minimum 10 Mbps broadband connection
- **Web Browser:** Google Chrome, Mozilla Firefox, Microsoft Edge (latest versions)

These hardware specifications ensure smooth performance, reliable system access, and efficient operation of the proposed College Admission Enquiry System.

2.4 SOFTWARE DESCRIPTION

2.4.1 Front End

The front end of the proposed College Admission Enquiry System is developed using Salesforce Lightning Web Components (LWC), a modern component-based framework used for building interactive user interfaces on the Salesforce platform. LWC is based on standard web technologies such as HTML, CSS, and JavaScript, and it allows developers to create reusable and efficient components. This framework ensures high performance, better user experience, and compatibility with modern web browsers.

Lightning Web Components enable the creation of responsive and user-friendly interfaces that allow students to interact with the admission system easily. The front end of the system includes several components that guide users through the admission process. These components provide functionalities such as checking seat availability in departments, verifying eligibility criteria, submitting admission enquiries, tracking application status, uploading required documents, and submitting fee payment details.

2.4.2 Back End

The back end of the proposed College Admission Enquiry System is developed using Salesforce Apex, a strongly typed and object-oriented programming language used for implementing server-side logic on the Salesforce platform. Apex runs on Salesforce's cloud infrastructure and acts as the connection between the user interface built with Lightning Web Components and the Salesforce database. It handles operations such as retrieving data, processing user inputs, validating information, and updating records in the system.

Apex enables the system to execute business logic required for the admission process, including managing student enquiries, storing document submissions, and processing fee payment details. It allows secure interaction with Salesforce objects using SOQL (Salesforce Object Query Language), which is used to retrieve and manipulate data stored in the database. Through Apex controllers, the front-end components can send requests and receive responses from the server, ensuring smooth communication between the user interface and the database.

2.4.3 Database

The database of the proposed College Admission Enquiry System is managed using the Salesforce cloud database, which is a secure and scalable multi-tenant relational database system. It stores all application data related to departments, student enquiries, document submissions, and fee payments. The database is accessed using SOQL (Salesforce Object Query Language) for retrieving records and Data Manipulation Language (DML) operations in Apex for inserting, updating, and deleting records.

The system uses several custom objects to organize and manage admission-related data. Objects such as Department, Admission Enquiry, Document Submission, and Fee Payment store information about available departments, student application details, submitted documents, and payment records. These objects are connected through relationships, allowing the system to link related data and maintain consistency across the database.

SUMMARY

This chapter presents a detailed analysis of the proposed College Admission Enquiry System by identifying the need to modernize the traditional manual admission process used in educational institutions. It includes a feasibility study that evaluates the system from economic, operational, and technical perspectives to ensure that the system can be developed and implemented effectively. The chapter also describes the required software and hardware specifications along with the technologies used for system development.

CHAPTER 3

SYSTEM DESIGN

3.1 MODULE DESCRIPTION

A module description provides a clear explanation of the purpose and functionality of each module within a system. It helps in understanding how different parts of the system work together to complete the overall process. In software development, modules divide the system into smaller functional units, making the application easier to develop, maintain, and manage. Each module performs a specific task and interacts with other modules to ensure smooth system operation.

The College Admission Enquiry System is organized into several modules that manage different stages of the admission process. These modules work together to provide a complete digital admission workflow for students and administrators.

The project consists of the following modules:

- Department and Seat Management Module
- Eligibility Checking Module
- Enquiry Submission Module
- Document Submission Module
- Fee Payment Module
- Reports and Dashboard Module

3.1.1 Department and Seat Management Module

The Department and Seat Management Module is responsible for managing the information related to academic departments and the availability of seats for admission. This module acts as the starting point of the admission process and provides the data required for other modules such as eligibility checking and enquiry submission. It stores details such as

department name, department code, programme type (UG or PG), total number of seats, available seats, annual fees, and minimum academic percentage requirements for admission.

This module allows administrators to create, update, and manage department records in the system. The information stored in this module helps students view the list of available departments and understand the eligibility criteria required for admission. The system also displays the current number of available seats for each department, helping students make informed decisions before applying.

3.1.2 Eligibility Checking Module

The Eligibility Checking Module is responsible for determining whether a student meets the minimum academic requirements for admission to a selected department. This module helps students verify their eligibility before submitting an admission enquiry, allowing them to understand whether they qualify for a particular programme based on their academic performance.

In this module, students select a department and enter their academic percentage marks such as 10th and 12th marks. If the selected programme is a postgraduate course, the system also requests the student's undergraduate percentage. The system then compares the entered marks with the minimum eligibility criteria defined for that department. This process helps students quickly identify whether they are eligible to apply for the selected course.

3.1.3 Enquiry Submission Module

The Enquiry Submission Module is responsible for collecting and storing the student's admission application details in the system. This module allows students to fill in an online form with their personal and academic information in order to apply for admission to a selected department. It acts as the formal step where a student submits their admission request to the institution.

In this module, students enter details such as their name, email address, mobile number, date of birth, and academic percentages for 10th, 12th, and undergraduate qualifications if required. They also select the department for which they want to apply. The system checks the

entered information to ensure that all required fields are filled correctly before the application is submitted.

3.1.4 Document Submission Module

The Document Submission Module allows students to submit the required documents needed to support their admission application. This module is available to students after they have successfully submitted their admission enquiry and logged into the system. Through the application tracking page, students can view their application details and indicate the documents they have submitted.

In this module, students are provided with a list of required documents such as academic certificates and other supporting records. They can mark the documents they have submitted and confirm the submission through the system. Once the documents are submitted, the information is stored.

3.1.5 Fee Payment Module

The Fee Payment Module is responsible for managing the payment of admission fees by students after their documents have been submitted and verified. This module allows students to provide payment details through an online form available in the system. The form collects important information such as payment mode, payment reference number, amount paid, payment date, and confirmation of receipt submission.

Once the student submits the payment information, the system records the payment details and links them to the corresponding admission enquiry. The admission status is then updated to indicate that the fee has been paid. This allows administration staff to easily identify applications that are ready for payment verification.

3.1.6 Reports and Dashboard Module

The Reporting and Dashboard Module provides administration staff with the tools required to manage and monitor the entire admission process. Through this module, administrators can review student enquiries, verify submitted documents, confirm fee payments,

and manage department details such as seat availability and eligibility criteria. This module helps staff oversee the admission workflow and ensure that all applications are processed correctly.

Administrators can access and manage records related to departments, admission enquiries, document submissions, and fee payments through the system interface. They can update department seat counts, verify student documents, and confirm payment details before finalizing admissions. The module allows staff to track the status of each application and manage the admission process efficiently.

3.2 DATA FLOW DIAGRAM

A Data Flow Diagram is a graphical representation that illustrates how data moves through a system. It shows how information flows between users, system processes, and data storage components. Data flow diagram help in understanding how input data is processed and transformed into output within the system. They also represent the interaction between external entities and the internal processes of the system. Data Flow Diagrams are commonly used during the system analysis and design phase to clearly visualize the structure and functionality of a system.

In the college admission enquiry system, the data flow diagram represents how information flows between students, administrators, and the admission management system. It shows how students provide information such as admission enquiries, academic details, document submissions, and fee payment details to the system, and how the system processes this information to generate responses such as eligibility results, application status updates, and admission confirmations.

3.2.1 DATA FLOW DIAGRAM LEVEL 0

Data flow diagram level 0, also known as the context diagram, represents the overall view of the college admission enquiry system as a single process. In this level, the entire system is represented as one main process called college admission enquiry system, and it shows how external entities interact with it.

In the diagram, the two main external entities are Student and Administrator. The student interacts with the system by providing inputs such as admission enquiry details, academic information, document submission, and fee payment details. The system processes these inputs and provides outputs such as eligibility results, application status updates, and admission confirmation messages.

This Level 0 data flow diagram provides a high-level overview of the system and defines the boundaries between external users and the College Admission Enquiry System before moving to more detailed in data flow diagram 1.

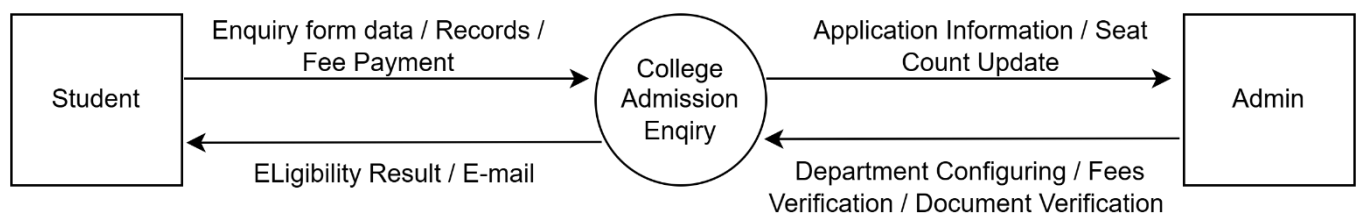


Figure 3.1 Data Flow Diagram Level 0

Figure 3.1, the data flow diagram Level 0 (Context Diagram) represents the overall view of the College Admission Enquiry System and shows how different external users interact with the system. In this diagram, the entire system is represented as a single process, and the interactions between the system and external entities such as Student and Administrator are shown through data flows.

In the system, students can access the admission portal to check department seat availability, verify their eligibility, submit admission enquiries, upload required documents, and provide fee payment details. The system processes the information provided by the students and updates their application status accordingly. Students can also track the progress of their application through the portal and receive notifications regarding document submission and admission confirmation.

The administrator has full access to the system and is responsible for managing department details, verifying submitted documents, confirming fee payments, and monitoring the admission process. The administrator also maintains department records such as available seats, eligibility criteria, and fee details.

3.2.2 DATA FLOW DIAGRAM LEVEL 1

Data Flow Diagram Level 1 provides a more detailed representation of the College Admission Enquiry System by breaking down the main process shown in data flow diagram level 0 into multiple smaller processes. It explains how the system internally handles different admission-related activities and how data flows between users, system processes, and data stores.

In this level, the system is divided into several processes like seat availability checking, application form submission, eligibility checking, document upload, document verification, fee payment, and record management. Students interact with the system by viewing available departments, filling out the admission application form, and submitting their academic details.

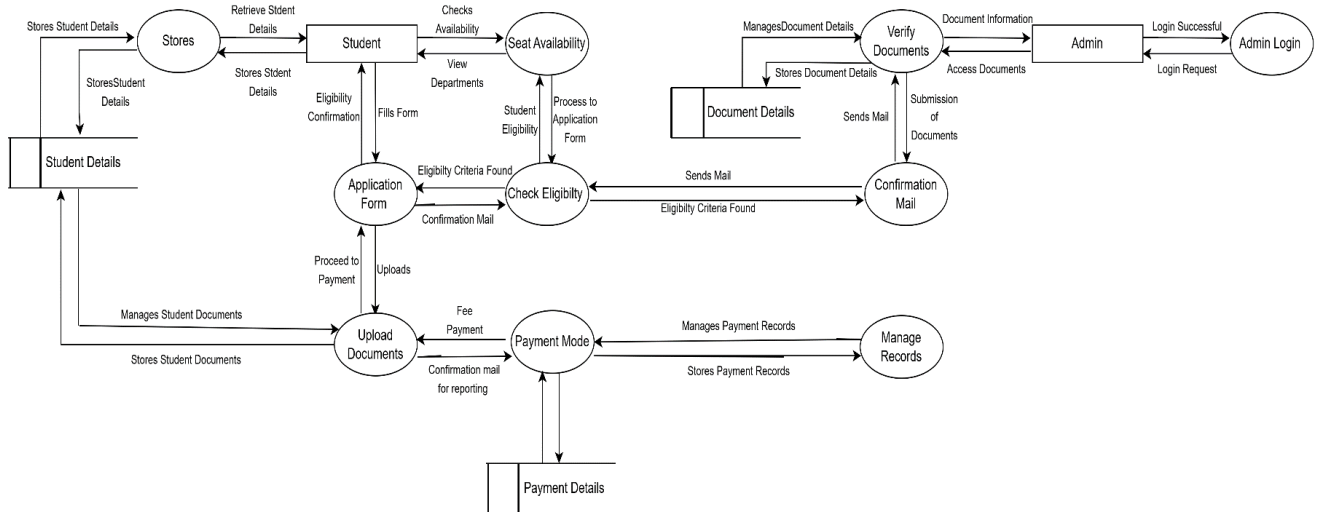


Figure 3.2 Data Flow Diagram Level 1

Figure 3.2, the data flow diagram level 1 diagram provides a detailed view of the college admission enquiry system by breaking down the main system into smaller processes and showing how data flows between users, processes, and data stores. The main users involved in this system are student and administrator.

The system stores important information such as student details, document details, and payment records in separate data stores. The administrator manages these records and monitors the admission process through the manage records process.

3.3 USE CASE DIAGRAM

A Use Case Diagram is a graphical representation used in system design to illustrate the interaction between users and the system. It shows the various functions or services provided by the system and how different users interact with them. The users who interact with the system are called actors, and the actions performed by them are known as use cases. Use case diagrams help in clearly understanding the system requirements and functionalities.

In the college admission enquiry system, the use case diagram includes two main actors: student and administrator. The Student interacts with the system to perform activities such as viewing department seat availability, checking eligibility criteria, filling out the application form, uploading required documents, and completing the admission fee payment.

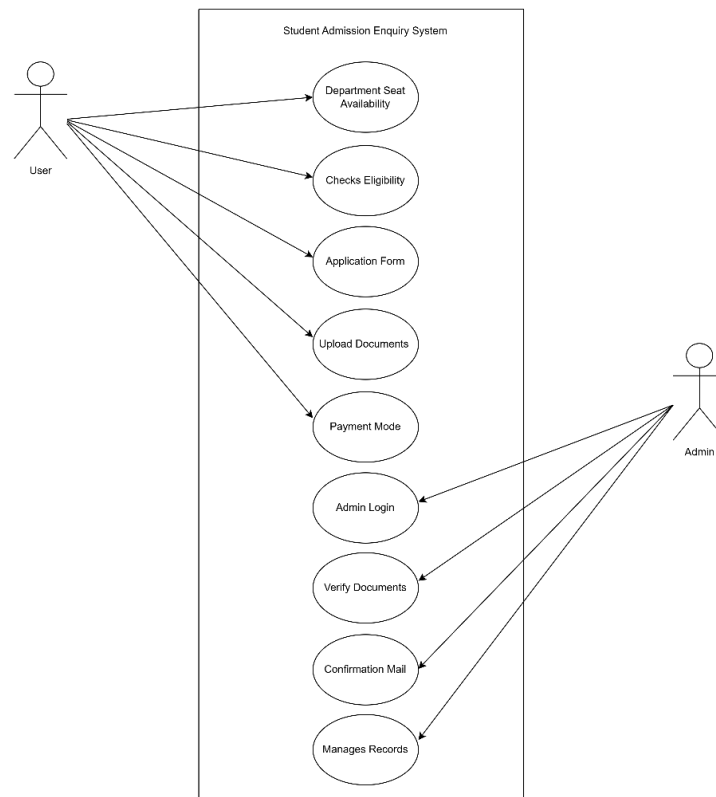


Figure 3.3 Use Case Diagram

Figure 3.3, the use case diagram of the college admission enquiry system represents the interaction between different users and the system functions. It shows how the main actors such as student and administrator interact with the system to perform various activities.

In this system, the student interacts with the portal to view department seat availability, check eligibility criteria, and fill out the admission application form. After submitting the application form, the student uploads the required documents and proceeds to the payment process to complete the admission enquiry. The student can also track the application status and receive updates regarding document verification and admission confirmation.

The administrator interacts with the system to manage department details, verify submitted documents, and review fee payment information. The admin is responsible for updating application status, confirming admissions, and ensuring that seat availability is maintained accurately. The system supports automated processes such as sending notifications and updating records, which helps reduce manual effort. Overall, the use case diagram explains how users interact with the system and how different processes are carried out efficiently

3.4 INPUT DESIGN

The input design refers to the process of creating user interfaces that allow users to enter data into the system in an accurate and efficient manner. A well-designed input interface helps reduce errors, improves usability, and ensures smooth interaction between users and the system. Effective input design focuses on simplicity, clarity, proper validation, and ease of use so that users can easily provide the required information without confusion. The input design also includes validation features that prevent incorrect or incomplete data from being entered into the system. Mandatory fields ensure that essential information such as student name, email address, department selection, and academic percentages are provided before submission, while data type validations help maintain accuracy.

In this system, input forms are designed for modules such as eligibility checking, admission enquiry, document submission, and fee payment. Each form includes clear labels, structured fields, and user-friendly layouts to guide users during data entry. Conditional fields are used where necessary, and proper error messages are displayed to help users correct mistakes easily. The system validates data at both client and server levels to ensure accuracy and security, which improves reliability and supports smooth functioning of the admission process.

Department Registration Form

New Department

* = Required Information

Information

* Department Name

Department Code

Programme Type

Total Seats

* Grade 10th Percentage

* Grade 12th Percentage

UG Percentage

Annual Fees

Description

Available Seats

Owner: JEVAMIRTHULA PON

Figure 3.4 Department Registration Form

Figure 3.4, the department registration form is used by the administrator to add and manage department details in the college admission enquiry system. The form allows the administrator to enter information such as department name, department code, programme type, total seats, available seats, annual fees, and minimum eligibility percentages. These details are stored in the system and are used by students when checking seat availability and eligibility.

The form ensures that all department information is entered accurately through validation rules such as unique department code and proper seat count limits. It provides options to create, update, or delete department records, making it easy for administrators to manage changes in department data. The stored information is dynamically displayed in the student portal, ensuring that students always view updated and correct details.

Eligibility Checker Form

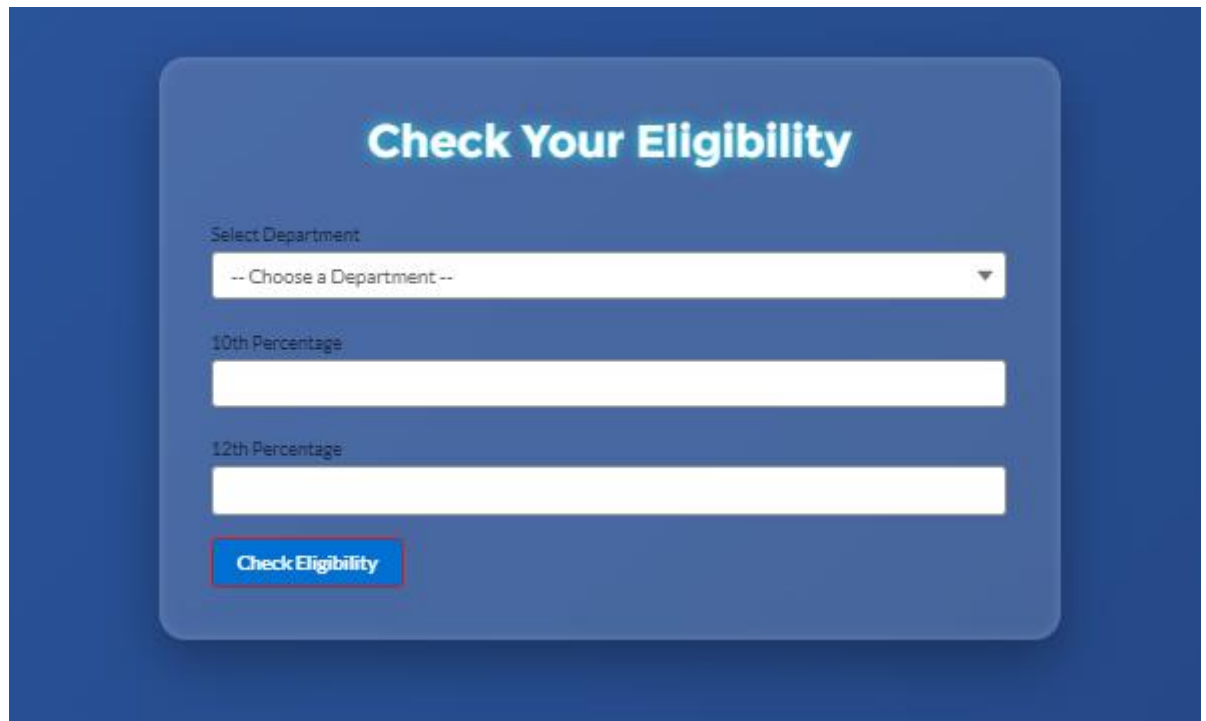
The image shows a web form titled "Check Your Eligibility" in white text on a dark blue background. The form itself is a lighter blue rounded rectangle. It contains a dropdown menu labeled "Select Department" with the placeholder text "-- Choose a Department --". Below this are two text input fields labeled "10th Percentage" and "12th Percentage". At the bottom of the form is a blue button with white text that says "Check Eligibility".

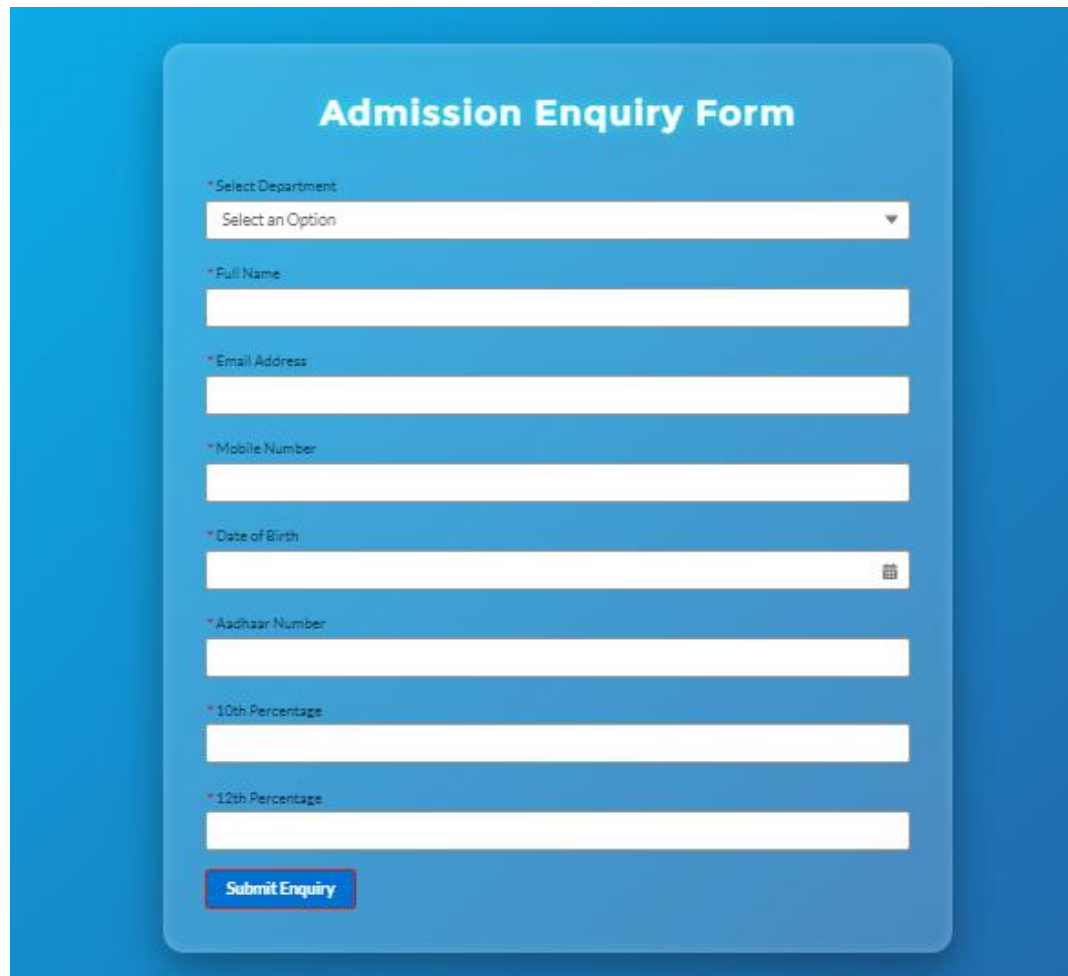
Figure 3.5 Eligibility Checker Form

Figure 3.5, the eligibility checker form allows students to check whether they meet the admission criteria for a selected department. In this form, students select a department and enter their 10th and 12th percentage marks, and if the programme is postgraduate, they must also enter their undergraduate percentage.

After submitting the form, the system compares the entered marks with the department's minimum eligibility criteria and displays the result instantly. If the student meets the requirements, they can proceed with the application process; otherwise, the system informs them that they are not eligible for that programme. The form provides quick feedback, helping students make informed decisions before applying.

The form is designed with simple input fields and validation checks to ensure that correct values are entered. It dynamically shows or hides fields based on the selected programme type, improving usability.

Admission Enquiry Form

The image shows a web form titled "Admission Enquiry Form" set against a blue background. The form itself is a lighter blue rounded rectangle. It contains several input fields, each preceded by a red asterisk indicating it is required. The fields are: "Select Department" (a dropdown menu with "Select an Option" and a downward arrow), "Full Name" (a text input), "Email Address" (a text input), "Mobile Number" (a text input), "Date of Birth" (a text input with a calendar icon on the right), "Aadhaar Number" (a text input), "10th Percentage" (a text input), and "12th Percentage" (a text input). At the bottom of the form is a blue button with white text that says "Submit Enquiry".

3.6 Admission Enquiry Form

Figure 3.6, the Admission Enquiry Form allows students to submit their application details in the College Admission Enquiry System. Through this form, students enter personal and academic information such as full name, email address, mobile number, date of birth, department selection and academic percentages. These details are stored in the system as an admission enquiry record.

The form includes validation checks to ensure that all required fields are filled correctly. For example, the email must follow a valid format, percentage values must be between 0 and 100, and the undergraduate percentage is required when applying for postgraduate programme.

Document Submission Form

The screenshot displays a web interface for document submission. At the top, a blue header contains the title "Application Status". Below this, a white box lists application details: "Enquiry No: ENQ-0117", "Department: Artificial Intelligence and Data Science", "Status: Doc Pending", and "Eligible: true". The main section, titled "Document Submission", features a list of required documents. Each document entry includes a label on the left and an upload area on the right. The upload area consists of a dashed box with an "Upload File" label, a red "Upload Files" button, and a dashed "Or drop files" label. The documents listed are: 10th Mark Certificate, 12th Mark Certificate, Transfer Certificate, Passport Size Photo, Government ID Proof, and UG Degree Certificate. A red "Submit Documents" button is located at the bottom of the form.

Document Type	Upload Action
10th Mark Certificate	Upload File (Upload Files button) Or drop files
12th Mark Certificate	Upload File (Upload Files button) Or drop files
Transfer Certificate	Upload File (Upload Files button) Or drop files
Passport Size Photo	Upload File (Upload Files button) Or drop files
Government ID Proof	Upload File (Upload Files button) Or drop files
UG Degree Certificate	Upload File (Upload Files button) Or drop files

Submit Documents

3.7 Document Submission Form

Figure 3.7, the Document Submission Form allows students to submit details of the documents required for the admission process. This form is available to authenticated students after they submit their admission enquiry and access the application tracking page. It displays important information such as enquiry number, student name, department, and eligibility status.

Students can select the documents they have submitted by checking the corresponding options such as 10th certificate, 12th certificate, transfer certificate, conduct certificate, passport photo, and identity proof.

Document Verification Form

Home Eligibility Check Enquiry Check Upload Doc Document Verification Admission Board																																																																																																																																																								
Admission Enquiries All New Printable View 51 items • Sorted by Enquiry Number • Filtered by All admission enquiries • Updated a few seconds ago <table> <tr> <th>☐</th><th>Enquiry Number</th><th>Student Name</th><th>Department</th><th>Date of Birth</th><th>10th Perc...</th><th>12th Perc...</th><th>UG Percen...</th><th>Owner First...</th></tr> <tr><td>1</td><td>ENQ-0001</td><td>Jeyamirthula</td><td>Compter Applications</td><td>7/15/2003</td><td>72</td><td>80</td><td>84</td><td></td></tr> <tr><td>2</td><td>ENQ-0002</td><td>JEYAMIRTHULA PON</td><td>Compter Applications</td><td>7/15/2003</td><td>72</td><td>72</td><td></td><td></td></tr> <tr><td>3</td><td>ENQ-0003</td><td>kowsalya</td><td>Compter Applications</td><td>11/15/2004</td><td>90</td><td>80</td><td></td><td></td></tr> <tr><td>4</td><td>ENQ-0032</td><td>Pramod Kumar Limma</td><td>Compter Applications</td><td>3/30/2011</td><td>89</td><td>89</td><td></td><td></td></tr> <tr><td>5</td><td>ENQ-0033</td><td>PKL</td><td>Compter Applications</td><td>3/12/2026</td><td>89</td><td>99</td><td></td><td></td></tr> <tr><td>6</td><td>ENQ-0073</td><td>leena</td><td>Compter Applications</td><td>3/15/2000</td><td>78</td><td>78</td><td></td><td></td></tr> <tr><td>7</td><td>ENQ-0074</td><td>priya</td><td>B.Tech-Food Technology</td><td>3/12/2002</td><td>76</td><td>76</td><td></td><td></td></tr> <tr><td>8</td><td>ENQ-0075</td><td>priya</td><td>B.Tech-Food Technology</td><td>3/12/2002</td><td>76</td><td>76</td><td></td><td></td></tr> <tr><td>9</td><td>ENQ-0076</td><td>priya</td><td>B.Tech-Food Technology</td><td>3/12/2002</td><td>76</td><td>76</td><td></td><td></td></tr> <tr><td>10</td><td>ENQ-0077</td><td>jeya</td><td>B.Tech-Food Technology</td><td>3/19/2001</td><td>80</td><td>80</td><td></td><td></td></tr> <tr><td>11</td><td>ENQ-0078</td><td>Gowtham</td><td>Compter Applications</td><td>4/14/2004</td><td>80</td><td>80</td><td></td><td></td></tr> <tr><td>12</td><td>ENQ-0079</td><td>kiruthi</td><td>B.Tech-Food Technology</td><td>3/17/2004</td><td>80</td><td>80</td><td></td><td></td></tr> <tr><td>13</td><td>ENQ-0080</td><td>tfty</td><td>Compter Applications</td><td>7/15/2003</td><td>90</td><td>90</td><td></td><td></td></tr> <tr><td>14</td><td>ENQ-0081</td><td>jeya</td><td>B.Tech-Food Technology</td><td>3/19/2003</td><td>80</td><td>80</td><td></td><td></td></tr> <tr><td>15</td><td>ENQ-0082</td><td>...</td><td>...</td><td>...</td><td>...</td><td>...</td><td></td><td></td></tr> </table>									☐	Enquiry Number	Student Name	Department	Date of Birth	10th Perc...	12th Perc...	UG Percen...	Owner First...	1	ENQ-0001	Jeyamirthula	Compter Applications	7/15/2003	72	80	84		2	ENQ-0002	JEYAMIRTHULA PON	Compter Applications	7/15/2003	72	72			3	ENQ-0003	kowsalya	Compter Applications	11/15/2004	90	80			4	ENQ-0032	Pramod Kumar Limma	Compter Applications	3/30/2011	89	89			5	ENQ-0033	PKL	Compter Applications	3/12/2026	89	99			6	ENQ-0073	leena	Compter Applications	3/15/2000	78	78			7	ENQ-0074	priya	B.Tech-Food Technology	3/12/2002	76	76			8	ENQ-0075	priya	B.Tech-Food Technology	3/12/2002	76	76			9	ENQ-0076	priya	B.Tech-Food Technology	3/12/2002	76	76			10	ENQ-0077	jeya	B.Tech-Food Technology	3/19/2001	80	80			11	ENQ-0078	Gowtham	Compter Applications	4/14/2004	80	80			12	ENQ-0079	kiruthi	B.Tech-Food Technology	3/17/2004	80	80			13	ENQ-0080	tfty	Compter Applications	7/15/2003	90	90			14	ENQ-0081	jeya	B.Tech-Food Technology	3/19/2003	80	80			15	ENQ-0082	...	...	...	...	...		
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Figure 3.8 Document Verification Form

Figure 3.8, the document verification form is used by the administrator to review and verify the documents submitted by students during the admission process. This form is available only to administrators through the salesforce internal interface and displays details such as document number, enquiry reference, document type, and submission date.

The administrator can verify each document by selecting the verification option and adding remarks if required. If a document is incorrect or incomplete, the administrator can reject it and provide a note explaining the reason. This process ensures that all submitted documents are carefully checked before proceeding to the next stage of the admission process.

Once the documents are verified, the system updates the application status accordingly, allowing the student to move to the next step such as fee payment. The form helps maintain accurate verification records and improves transparency in the admission process. It reduces manual tracking and ensures that document verification is completed efficiently and systematically.

Fee Payment Details Module

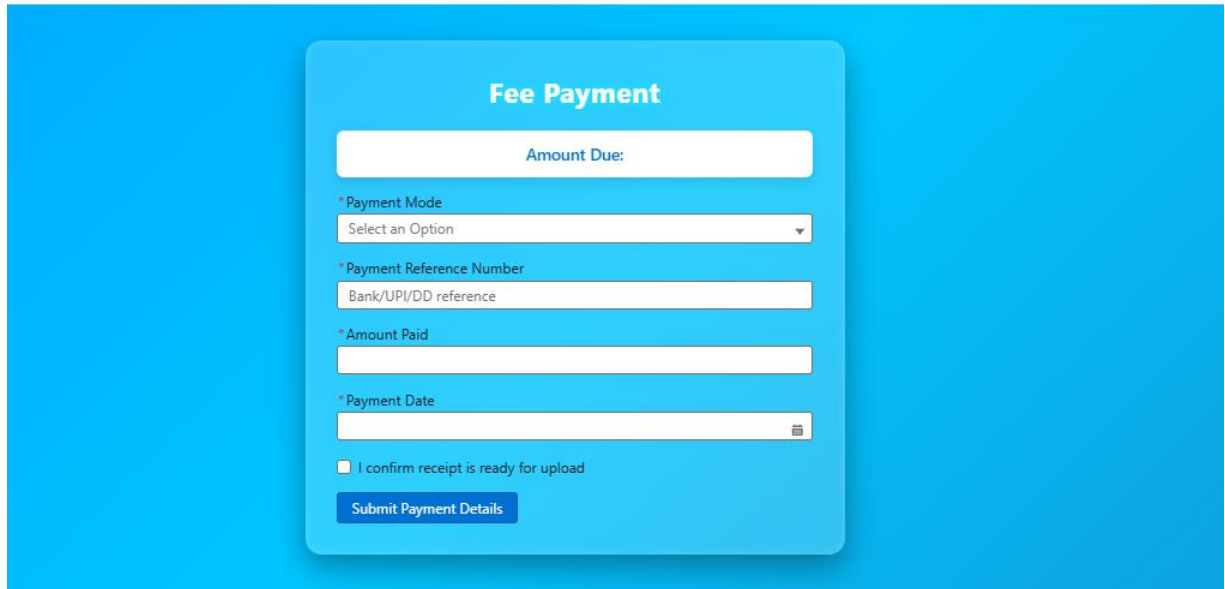
The image shows a web form titled "Fee Payment" on a blue background. The form is white with a blue border. It contains several input fields: a text field for "Amount Due:", a dropdown menu for "Payment Mode" with the placeholder "Select an Option", a text field for "Payment Reference Number" with the placeholder "Bank/UPI/DD reference", a text field for "Amount Paid", and a date picker for "Payment Date". Below these fields is a checkbox labeled "I confirm receipt is ready for upload" and a blue button labeled "Submit Payment Details".

Figure 3.9 Fee Payment Details

Figure 3.9, the Fee Payment Module allows students to submit their admission fee details after completing the document verification stage. This module is accessible to authenticated students through the portal and displays important information such as the amount to be paid based on the selected department. It provides a structured form where students can enter payment details including payment mode, payment reference number, amount paid, and payment date.

The module includes validation checks to ensure that all required fields are filled correctly. Students must select a valid payment mode such as online transfer, bank transfer, cash, cheque, or demand draft, and provide a valid transaction reference number. The amount entered must match the required fee, and the payment date must be valid. An option is also provided for students to confirm that the payment receipt is ready for upload.

After submitting the payment details, the information is stored in the system and linked to the corresponding admission enquiry. Administrators can then review and verify the payment through the system. Once the payment is verified, the admission status is updated, and the student receives confirmation of seat allocation.

3.5 OUTPUT DESIGN

Output Design refers to the process of presenting the results produced by a system to the users in a clear and understandable format. It focuses on how the processed data is displayed to the users through reports, messages or screens. The main objective of output design is to ensure that the information generated by the system is accurate, meaningful and easy to interpret. A good output design helps users quickly understand the results and supports effective decision-making.

Enquiry Submission Confirmation Screen

The system automatically generates a unique enquiry number for each submitted application, which the student can use to track the progress of their admission. The screen also informs the student that a document request email has been sent to their registered email address. This output design helps provide clear confirmation to the student and ensures that they are informed about the next step in the admission process.

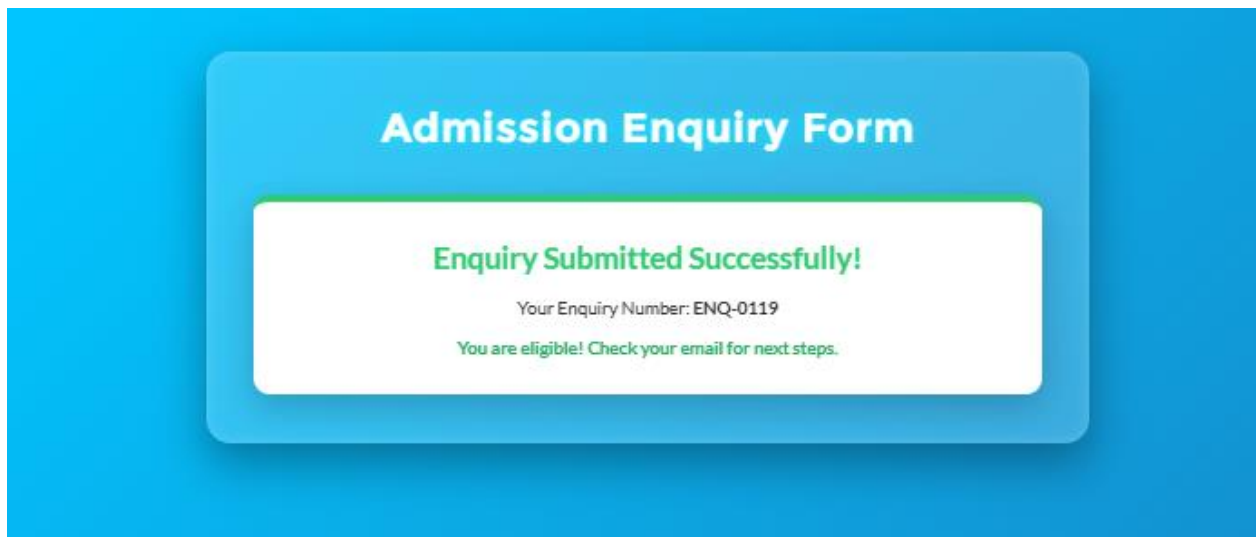


Figure 3.10 Enquiry Submission Confirmation Screen

Figure 3.10, the enquiry submission confirmation screen is displayed after a student successfully submits the admission enquiry form. This output screen confirms that the student's application has been received by the system. It shows important details such as the enquiry number, student name, selected department, application date, and eligibility status.

Eligibility Result Screen

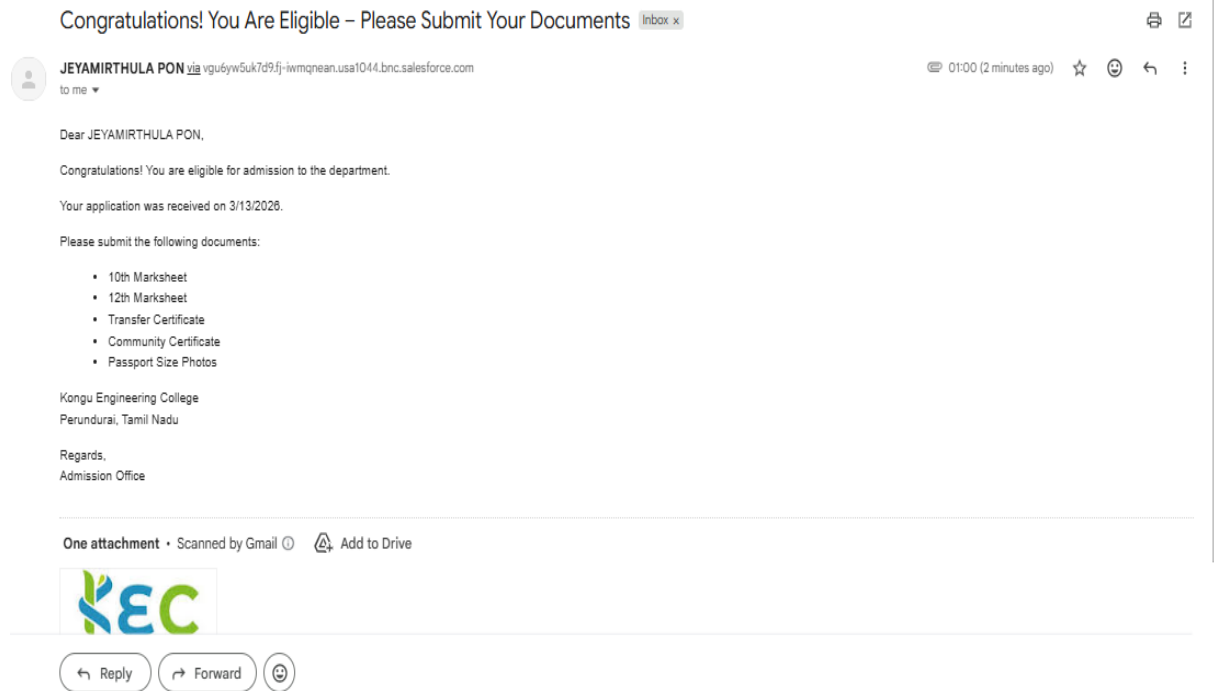


Figure 3.11 Eligibility Result Screen

Figure 3.11, the eligibility result screen is displayed after a student clicks the check eligibility button in the eligibility checker form. This output screen shows whether the student is eligible for the selected department based on the entered academic percentage marks.

The screen displays important information such as the department name, programme type, the student's marks, and the minimum required marks for eligibility. It also shows whether the student has passed each eligibility criterion. In addition, the screen provides details such as the annual fee for the selected department and the number of seats currently available.

If the student is eligible, the screen provides an option to proceed to the admission enquiry form, allowing them to continue the application process. If the student is not eligible, the system clearly indicates the reason, helping the student understand which criteria were not met. This screen improves transparency and helps students make informed decisions before applying.

Seat Availability Display

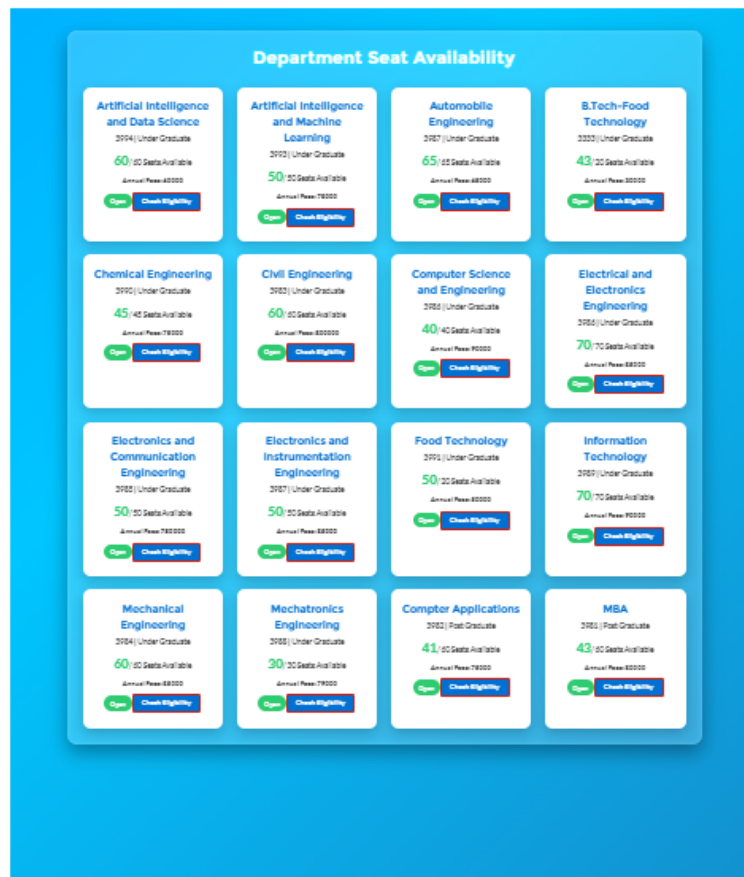


Figure 3.12 Seat Availability Display

Figure 3.12, the seat availability display screen shows the list of departments available for admission when the portal page loads. This output screen allows students to view important information about each department such as programme type, available seats, total seats, and annual fees.

Each department is displayed in a card format showing the current seat availability and whether the department is open or full for admission. A visual indicator also represents the percentage of seats filled in each department. If seats are available, students can select the option to check eligibility for that department. The data displayed on this screen is updated in real time, ensuring that students always see accurate seat availability

Application Status Tracker Screen

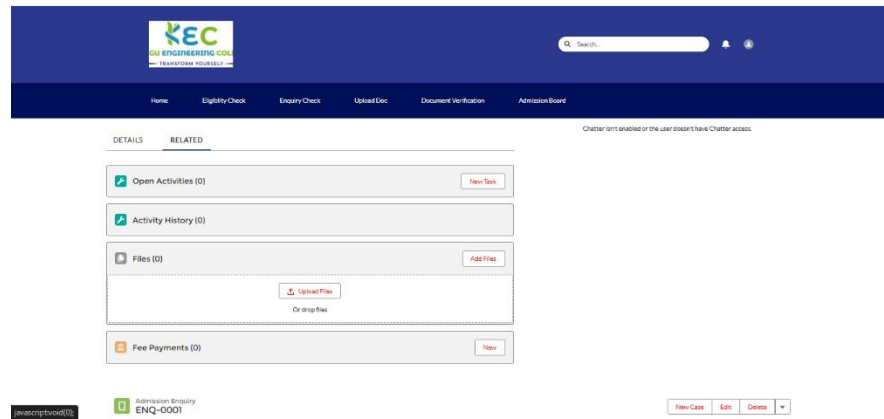


Figure 3.13 Application Status Tracker Screen

Figure 3.13, the Application Status Tracker Screen allows students to view the progress of their admission application. It displays details such as enquiry number, student name, department, application date, and current admission status. The screen also shows the stages of the admission process including enquiry submission, eligibility check, document submission, fee payment, and confirmation. Students can view the documents they have submitted and check whether they are verified or pending.

The tracker provides real-time updates, helping students stay informed about each step of their application without contacting the administration. It clearly indicates the current stage and guides students on the next action to be completed. This screen improves transparency, reduces communication delays, and makes it easier for students to monitor their admission progress efficiently.

It ensures that all application details are displayed in a clear and organized manner, making it easy for students to understand their status. Overall, it enhances user experience and supports smooth coordination between students and administrators throughout the admission process.

Student Admission Dashboard

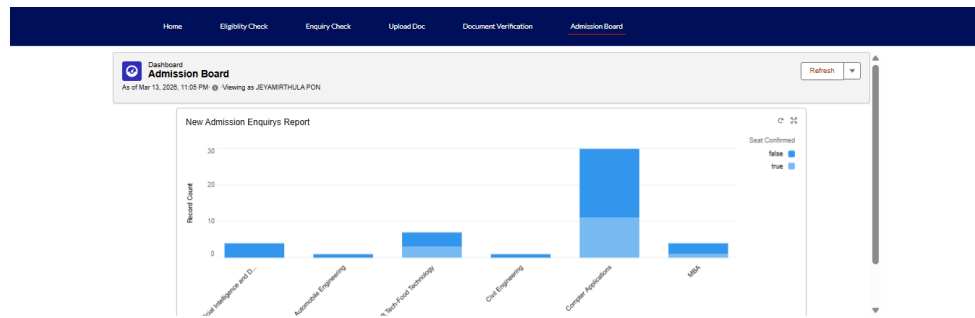


Figure 3.14 Student Admission Dashboard

Figure 3.14, the student admission dashboard allows administrators to monitor the overall admission process. It displays key information such as total enquiries, eligible students, documents submitted, and confirmed admissions. The dashboard also includes charts and reports showing seat availability by department and admission status. These reports help administrators track admission progress and manage the admission process effectively.

The dashboard provides real-time data updates, enabling administrators to make quick and informed decisions. It helps identify departments with high or low admissions and monitor each stage of the admission pipeline. The visual representation of data improves understanding and reduces the time required to analyze information.

In addition, the dashboard supports better planning and resource management by providing a clear overview of the admission process. It enhances transparency, improves efficiency, and ensures that administrators can manage admissions smoothly and accurately. The system allows easy filtering and viewing of records based on different criteria. It reduces manual reporting effort and improves data accuracy. It also helps administrators quickly respond to changes during the admission process.

3.6 ER DIAGRAM

An Entity Relationship Diagram (Entity Relationship Diagram) is a graphical representation used to show the structure of a database. It illustrates the entities present in a system and the relationships between them. Entities represent real-world objects such as students, mentors, or reports. The diagram also shows attributes that describe each entity. ER diagrams help in designing and organizing the database structure of a system.

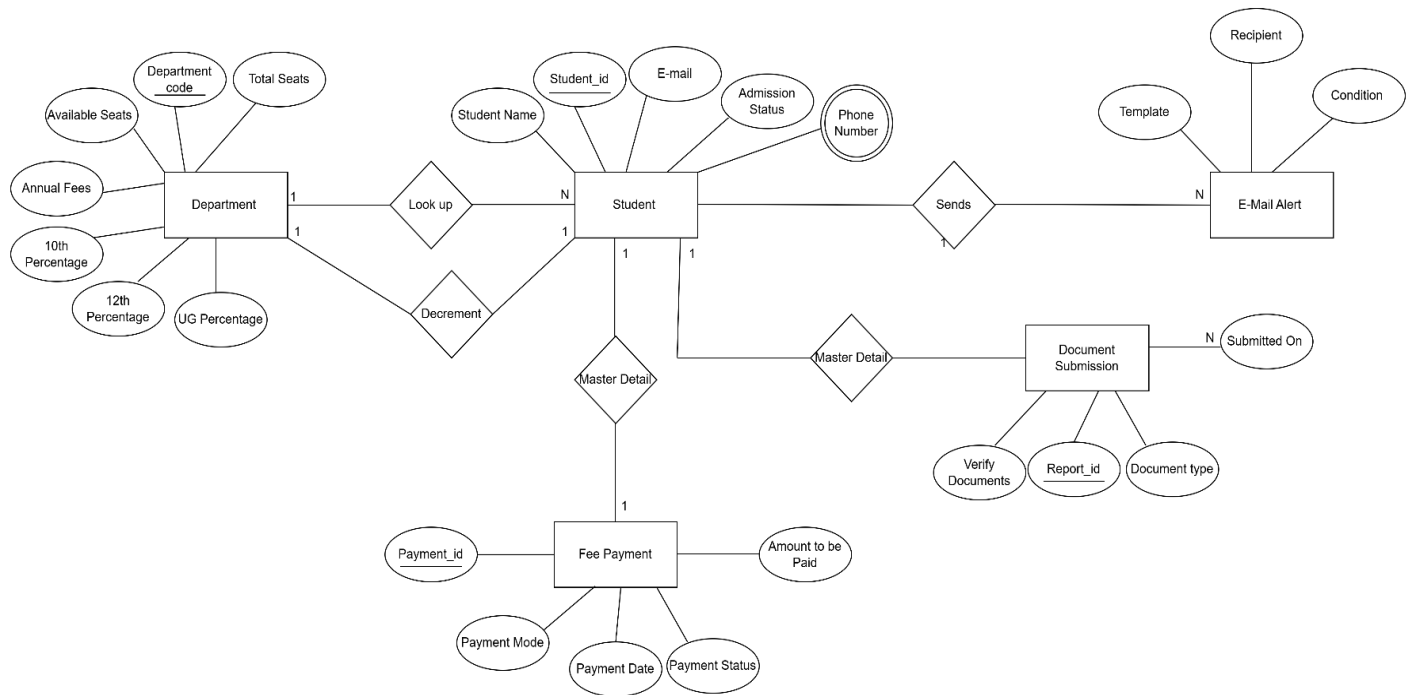


Figure 3.15 ER Diagram

The entity relationship diagram of the college admission enquiry system shows the relationships between the main entities involved in the system such as department, student, document submission, fee payment, and e-mail alert. These entities represent the core components of the admission process and help organize the data related to student applications, department information, documents, and payment details. The diagram illustrates how student information, department seat availability, admission status, and payment records are stored and managed within the system.

3.7 DATABASE DESIGN

Database design refers to the process of organizing and structuring data so that it can be stored, managed and retrieved efficiently. It defines how information is stored in the system and how different data elements are related to each other. A well-designed database ensures data accuracy, consistency and easy access to information. Proper database design also helps reduce data redundancy and improves overall system performance.

In the College Admission Enquiry System the database is implemented using the Salesforce platform, where data is stored in the form of custom objects, fields and relationships. Objects act similar to database tables and store records related to different entities such as departments, student enquiries, document submissions and fee payments. Fields store specific information such as student name, email, phone number, academic percentages, admission status and payment details. This structure helps organize admission data in a systematic and efficient way.

3.8 TABLE DESIGN

Table Design refers to the process of organizing data into structured tables in a database. Each table consists of rows and columns where rows represent records and columns represent fields. Table design defines the field names, data types and relationships between different tables. A well-designed table helps store and retrieve data efficiently. It also ensures data accuracy, consistency and easy management of information in the system.

The table design is implemented using Salesforce objects where each object represents a table and fields represent columns. Proper relationships such as lookup and master-detail are used to connect different tables like department, admission enquiry, document submission, and fee payment. This helps maintain data integrity and ensures that related records are properly linked.

The design includes validation rules and constraints to prevent incorrect or duplicate data entry. It supports efficient data retrieval for reports and dashboards, enabling administrators to analyze admission data easily. The structured design reduces data redundancy and improves system performance.

3.8.1 DEPARTMENT TABLE

Table 3.1 Department Table

Field Name	Data Type	Description
Name	Text	Name of the department
Department Code	Text	Unique code assigned to the department
Programme Type	Picklist	Type of programme such as UG or PG
Total Seats	Number	Total number of seats available in the department
Available Seats	Number	Number of seats currently available for admission
Annual Fees	Currency	Annual fee amount for the department
Min 10th Percentage	Number	Minimum 10th percentage required for eligibility
Min 12th Percentage	Number	Minimum 12th percentage required for eligibility
Min UG Percentage	Number	Minimum UG percentage required for PG programmes

In **Table 3.1**, the Department table stores information about all departments available in the College Admission Enquiry System. The Department Code acts as the primary key to uniquely identify each department record. This table contains details such as department name, department code, programme type, seat availability, annual fees, and minimum eligibility criteria required for admission. No foreign key is used in this table because it represents the master data for departments.

Additionally, this table helps the system display department details and seat availability to students during the admission enquiry process. It is also used to validate student eligibility based on minimum percentage requirements before submitting an admission enquiry.

3.8.2 ADMISSION ENQUIRY TABLE

Table 3.2 Admission Enquiry Table

Field Name	Data Type	Description
Enquiry Number	Auto Number	Unique enquiry number generated automatically in the format ENQ-0001
Department	Lookup	Reference to the selected department
Student Name	Text	Full name of the student applying for admission
Email Address	Email	Email address of the applicant
Mobile Number	Phone	Contact number of the student
Date of Birth	Date	Birth date of the student
10th Percentage	Number	Percentage obtained in the 10th examination
12th Percentage	Number	Percentage obtained in the 12th examination
UG Percentage	Number	Undergraduate percentage required for PG programmes
Is Eligible	Formula (Boolean)	Automatically calculated eligibility status based on marks
Admission Status	Picklist	Current application status such as New, Document Pending, Document Submitted, Fee Paid or Confirmed

Document Email Sent	Checkbox	Indicates whether the document request email has been sent
Fee Payment Status	Picklist	Shows payment status such as Not Paid, Submitted or Verified
Seat Confirmed	Checkbox	Indicates whether the admission seat is confirmed
Confirmation Email Sent	Checkbox	Indicates whether the confirmation email has been sent
Application Date	Date	Date when the enquiry was submitted

In **Table 3.2**, the admission enquiry table stores details of student admission applications in the college admission enquiry system. The enquiry number acts as the primary key, while the department field acts as a foreign key referencing **Table 3.1**. It contains student personal information, academic percentages, department selection, and admission status. The table also records eligibility results, document submission status, fee payment status, and seat confirmation details to track the admission process.

3.8.3 DOCUMENT SUBMISSION TABLE

Table 3.3 Document Submission Table

Field Name	Data Type	Description
Document Number	Auto Number	Unique document record number generated automatically (DOC-0001)
Admission Enquiry	Master-Detail	Reference to the related admission enquiry
Document Type	Picklist	Type of document submitted

		such as 10th Certificate, 12th Certificate, Transfer Certificate, Passport Photo, etc.
Submitted On	Date	Date on which the document was submitted
Verified	Checkbox	Indicates whether the document has been verified by the administrator
Notes	Text Area	Remarks or comments added by the administrator during verification

In **Table 3.3**, the document submission table stores details of documents submitted by students for admission verification. The document number acts as the primary key, while the admission enquiry field acts as a foreign key referencing **Table 3.2**. This table records document type, submission date, and verification status. It helps administrators verify student documents and track the document submission process in the college admission enquiry system.

SUMMARY

This chapter explained the system design of the College Admission Enquiry System by presenting important design diagrams and database structures used in the system. It included diagrams such as the Data Flow Diagram, Use Case Diagram, and Entity Relationship Diagram which describe how users interact with the system and how data moves between different modules. These diagrams help in clearly understanding the structure, workflow, and functionality of the system.

CHAPTER 4

IMPLEMENTATION

4.1 CODE DESCRIPTION

Code description is important for explaining the functionality and structure of the source code used in the College Admission Enquiry System. The project is implemented using the Salesforce platform, which includes Lightning Web Components for the frontend interface, Apex for backend logic, and Salesforce objects and automation tools for managing data and workflows. Each module of the system is developed carefully to ensure clarity, modularity, and easy maintenance.

The code structure follows the standard Salesforce DX project format, where components and classes are organized into specific folders. Lightning Web Components are used to create the user interface for features such as seat availability, eligibility checking, enquiry submission, document submission, and fee payment. Each component contains separate files for HTML markup, JavaScript logic, and configuration metadata. Apex classes handle server-side operations such as retrieving department data, processing enquiries, storing document submissions, and recording payment details.

4.2 STANDARDIZATION OF THE CODING

Coding standards are strictly followed throughout the development of the College Admission Enquiry System to maintain consistency, readability, and scalability of the application. Since the project is developed using the Salesforce platform, standard development and configuration practices are applied while creating custom objects, fields, Lightning Web Components, Apex classes, and automation flows. Meaningful naming conventions are used for objects such as department, admission enquiry, document submission, and fee payment so that each component of the system can be easily identified and managed.

Following standardized development practices also helps ensure that the system remains reliable and easy to maintain in the future. Lightning web components are organized with clear

separation of HTML, JavaScript, and metadata files, while apex classes are written using secure coding practices and clear method structures. Salesforce flows are designed in a simple and modular way to automate tasks such as sending document request emails, confirming admissions, and updating seat availability.

4.3 ERROR HANDLING

Error handling is an important part of the college admission enquiry system to ensure that the system operates smoothly and prevents incorrect or incomplete data from being stored. The system includes validation checks and proper field constraints to ensure that users enter correct and complete information while submitting admission enquiries, documents, and fee payment details. Required fields such as student name, email, department selection, and academic percentages must be completed before submitting the form, which helps reduce data entry mistakes.

Salesforce validation rules and appropriate field data types are used to prevent invalid data from being saved in the system. For example, email fields accept only valid email formats and numeric fields ensure that percentage values are entered correctly. In addition, Salesforce flows and automation processes are configured to follow the correct admission workflow when sending document request emails, confirming admissions, and updating seat availability.

SUMMARY

This chapter explained the implementation aspects of the College Admission Enquiry System developed using the Salesforce platform. It described the coding approach used for Lightning Web Components and Apex classes, along with the structure and organization of the system components. The chapter also discussed the coding standards followed during development to ensure consistency, readability, and maintainability of the application.

CHAPTER 5

TESTING AND RESULTS

5.1 Testing

Testing is an important phase in the development of the College Admission Enquiry System to ensure that the system functions correctly and meets the required objectives. It involves verifying that all modules such as department management, eligibility checking, enquiry submission, document submission, fee payment, and reporting are working properly without errors. Testing helps identify problems in the system and ensures that each component performs its intended function effectively.

In this project, testing is performed to verify whether students can check seat availability, confirm eligibility, submit admission enquiries, upload documents, and complete fee payment through the system. It also ensures that the information entered by users is stored correctly in the Salesforce database and that administrators can view accurate data through reports and dashboards. Through proper testing, the reliability, accuracy, and performance of the system are ensured before the system is fully implemented. The testing techniques applied in this project include:

- Unit Testing
- Integration Testing
- User Acceptance Testing

5.1.1 UNIT TESTING

Unit testing is performed to test individual modules of the system separately. Each module such as department and seat management, eligibility checking, enquiry submission, document submission, and fee payment is tested individually to ensure that it performs its intended function correctly. This type of testing helps identify errors within specific components before integrating them with other modules.

CASE 1

Module: Department and Seat Management Module

Test Type: Department Data Display Verification

Input: Department Name, Programme Type, Total Seats, Available Seats

Output: Department details displayed with correct seat availability

TEST

Input: Administrator adds department details in the system

Output: Department information appears correctly in the seat availability page

Analysis: The department information is stored correctly and displayed with the correct seat availability.

Result: Pass

CASE 2

Module: Eligibility Checking Module

Test Type: Eligibility Verification

Input: Department selection with student percentage marks

Output: Eligibility status displayed (Eligible / Not Eligible)

TEST

Input: Student enters valid academic percentage values

Output: System calculates eligibility based on department criteria

Analysis: The system correctly evaluates eligibility based on the defined percentage requirements.

Result: Pass

CASE 3

Module: Enquiry Submission Module

Test Type: Admission Enquiry Creation

Input: Student Name, Email, Mobile Number, Department Selection, Academic Marks

Output: Admission Enquiry Record Created

TEST

Input: Student submits valid enquiry details

Output: Enquiry record created in Admission Enquiry object with unique enquiry number.

Analysis: The enquiry data is saved correctly and linked to the selected department.

Result: Pass

CASE 4

Module: Document Submission Module

Test Type: Document Submission Verification

Input: Selected document types for submission

Output: Document submission records created and linked to the enquiry

TEST

Input: Student selects and submits required document types

Output: Document records stored in Document Submission object

Analysis: The submitted documents are correctly recorded and associated with the admission enquiry.

Result: Pass

CASE 5

Module: Fee Payment Module

Test Type: Fee Payment Verification

Input: Payment Mode, Payment Reference, Amount Paid, Payment Date

Output: Fee Payment Record Created

TEST

Input: Student submits valid payment details

Output: Fee payment record stored and admission status updated

Analysis: Payment information is recorded correctly and linked with the admission enquiry record.

Result: Pass

CASE 6

Module: Administratorisation and Reporting Module

Test Type: Report Generation Verification

Input: Admission enquiry and payment data stored in the system

TEST

Input: Administrator opens admission reports and dashboards

Output: Reports display seat availability, admission status, and payment details

Analysis: The reports correctly retrieve data from the system and display accurate admission statistics.

Result: Pass

5.1.2 INTEGRATION TESTING

Integration testing is used to verify that different modules of the College Admission Enquiry System work together correctly. In this project, it ensures that once a student submits an enquiry, the system correctly triggers the document request email flow and allows the student to proceed with document submission and fee payment. This testing confirms that the interaction between modules functions smoothly and that data flows correctly between modules such as department management, enquiry submission, document submission, fee payment, and reporting. Integration testing also verifies that admission details are correctly stored in the Salesforce database and updated across different stages of the admission process.

CASE 1

Modules: Department and Seat Management Module → Enquiry Submission Module

Test Type: Module Interaction Verification

Input: Student selects a department and submits an enquiry

TEST

Input: Valid enquiry details submitted

Output: Admission enquiry record created and linked to the selected department

Analysis: The enquiry submitted by the student is successfully stored and connected with the department record.

Result: Pass

CASE 2

Modules: Enquiry Submission Module → Document Submission Module

Test Type: Automation Trigger Verification

Input: Student submits a valid admission enquiry

TEST

Input: Enquiry record created with eligible status

Output: Document request email is triggered automatically

Analysis: The system correctly sends a document submission request email after a valid enquiry is created.

Result: Pass

CASE 3

Modules: Document Submission Module → Fee Payment Module

Test Type: Workflow Progress Verification

Input: Student submits required documents

TEST

Input: Documents recorded in the system

Output: Student can proceed to the fee payment stage

Analysis: The system correctly records document submissions and allows the next step of fee payment.

Result: Pass

CASE 4

Modules: Fee Payment Module → Department and Seat Management Module

Test Type: Seat Update Verification

Input: Fee payment verified by administrator

TEST

Input: Payment status updated to verified

Output: Admission confirmed and available seat count reduced

Analysis: The system correctly updates the department seat availability after confirming admission.

Result: Pass

CASE 5

Modules: Admission Records → Reports and Dashboard Module

Test Type: Data Retrieval Verification

Input: Admission enquiry and payment data stored in the system

TEST

Input: Administrator opens admission reports

Output: Reports display admission status, seat availability, and payment details

Analysis: The reports correctly retrieve and display stored admission data from the system.

Result: Pass

5.1.3 VALIDATION TESTING

Validation testing is performed to ensure that the College Admission Enquiry System correctly follows the defined business rules and satisfies the requirements specified in the system design. This testing verifies that the system performs the correct operations according to user needs and institutional policies. Validation testing focuses on checking system rules such as eligibility criteria, seat availability restrictions, email format validation, and correct admission workflow processing.

CASE 1

Module: Enquiry Submission Module

Test Type: Seat Availability Validation

Input: Department with zero available seats

TEST

Input: Student attempts to submit enquiry

Output: System blocks enquiry submission and displays message

Analysis: The system correctly prevents application submission when no seats are available.

Result: Pass

CASE 2

Module: Eligibility Checking Module

Test Type: PG Eligibility Validation

Input: Postgraduate department selected without UG percentage

TEST

Input: Student submits enquiry without entering UG marks

Output: System displays error requiring UG percentage

Analysis: The validation rule correctly enforces the requirement for undergraduate marks in postgraduate programmes.

Result: Pass

CASE 3

Module: Enquiry Submission Module

Test Type: Email Format Validation

Input: Invalid email format entered

TEST

Input: Email value such as "invalidemail"

Output: System displays email format error

Analysis: The system correctly validates email format and prevents invalid data from being stored.

Result: Pass

CASE 4

Module: Eligibility Checking Module

Test Type: Boundary Value Verification

Input: Student marks equal to minimum eligibility value

TEST

Input: Marks exactly equal to required minimum percentage

Output: Student marked as eligible

Analysis: The system correctly treats boundary values as valid when they meet the minimum criteria.

Result: Pass

CASE 5

Module: Fee Payment Module

Test Type: Admission Confirmation Verification

Input: Fee payment verified by administrator

TEST

Input: Payment status updated to verified

Output: Seat confirmation email sent and seat count updated

Analysis: The system correctly confirms admission and updates seat availability.

Result: Pass

CASE 6**Module:** Administratoristration and Reporting Module**Test Type:** Dashboard Update Verification**Input:** Admission confirmation completed**TEST****Input:** Administrator opens admission dashboard**Output:** Dashboard reflects updated admission and seat data**Analysis:** Reports and dashboards correctly display real-time admission information.**Result:** Pass**SUMMARY**

This chapter discussed the testing and results of the College Admission Enquiry System to ensure that the system functions correctly and meets the specified requirements. Different testing methods such as unit testing, integration testing, and validation testing were performed to verify the functionality of individual modules and the interaction between system components. These tests ensured that features such as department management, eligibility checking, enquiry submission, document submission, fee payment, and reporting work accurately without errors.

The testing process confirmed that the system correctly follows business rules, handles user inputs properly, and updates data consistently across the system. The results show that all modules operate as expected and the automated workflows function correctly.

CHAPTER 6

CONCLUSION AND FUTURE ENHANCEMENTS

6.1 Conclusion

The College Admission Enquiry System developed using the Salesforce platform provides an automated solution for managing the admission enquiry and application process in educational institutions. It uses Salesforce technologies such as Lightning Web Components, Apex, Experience Cloud, Flows, Reports, and Dashboards to replace the traditional manual admission process with a centralized digital system. The system allows students to check seat availability, verify eligibility, submit admission enquiries, upload documents, and complete fee payments through an online platform.

Administrators can manage department details, verify documents, confirm fee payments, and monitor admission activities using reports and dashboards. Automated features such as eligibility checking, email notifications, and seat updates help streamline the admission workflow and reduce manual effort.

6.2 Future Enhancement

The College Admission Enquiry System can be improved in the future by adding features such as online document upload and secure online fee payment to make the admission process fully digital. The system can also include automated merit list generation, waitlist management, and mobile application support to improve accessibility for students. Further enhancements may include AI-based course recommendations, multi-language support, and integration with other institutional systems to simplify data management.

APPENDICES

A. SAMPLE CODING

HOME PAGE

AdmissionController.cls

```
public without sharing class AdmissionController {

    @AuraEnabled(cacheable=true)

    public static List<Department__c> getDepartments() {

        return [

            SELECT Id,Name,Department_Code__c,Programme_Type__c, Total_Seats__c,
            Available_Seats__c, Annual_Fees__c, Description__c, Grade_10th_Percentage__c,
            Grade_12th_Percentage__c, UG_Percentage__c FROM Department__c WITH

        ]; }    @AuraEnabled

    public static Admission_Enquiry__c searchEnquiry(String enquiryNumber, String
aadharNumber){ try{

List<Admission_Enquiry__c> enq = [SELECT Id, Name, Aadhar_Number__c,
Is_Eligible__c, Admission_Status__c, Fee_Payment_Status__c, Seat_Confirmed__c,
Department__c, Department__r.Name, FROM Admission_Enquiry__c WHERE Name =
:enquiryNumber AND Aadhar_Number__c = :aadharNumber LIMIT 1];

if(enq.isEmpty()){

        throw new AuraHandledException('Invalid Enquiry Number or Aadhar Number');

return enq[0];
```

```

    }catch(Exception e){

throw new AuraHandledException('Search failed: ' + e.getMessage());

}}

```

CHECK ELIGIBILITY PAGE

eligibilityChecker.js

```

import { LightningElement, wire, track } from 'lwc';
import { NavigationMixin } from 'lightning/navigation';
import getDepartments from '@salesforce/apex/AdmissionController.getDepartments';

export default class EligibilityChecker extends NavigationMixin(LightningElement) {
    @track selectedDeptId = "";
    @track tenth = null;
    @track twelfth = null;
    @track ugPercent = null;
    @track resultShown = false;
    @track isEligible = false;
    @track resultMessage = "";
    @track isPG = false;
    deptOptions = [];
    allDepts = [];

    @wire(getDepartments)
    wiredDepts({ data }) {
        if (data) {
            this.allDepts = data;
            this.deptOptions = data.map(d => ({
                label: d.Name + ' (' + d.Programme_Type__c + ')',
                value: d.Id
            }));
        }
    }
}

```

```

    }));
  }
}

```

```

handleDeptChange(e) {
  this.selectedDeptId = e.detail.value;
  const dept = this.allDepts.find(d => d.Id === this.selectedDeptId);
  this.isPG = dept && dept.Programme_Type__c === 'PG';
  this.resultShown = false;
}

```

```

handleTenth(e) { this.tenth = parseFloat(e.detail.value); }
handleTwelfth(e) { this.twelfth = parseFloat(e.detail.value); }
handleUG(e) { this.ugPercent = parseFloat(e.detail.value); }

```

```

checkEligibility() {
  const dept = this.allDepts.find(d => d.Id === this.selectedDeptId);
  if (!dept) return;
  let eligible;
  if (dept.Programme_Type__c === 'PG') {
    eligible = this.tenth >= dept.Grade_10th_Percentage__c &&
      this.twelfth >= dept.Grade_12th_Percentage__c &&
      this.ugPercent >= dept.UG_Percentage__c;
  } else {
    eligible = this.tenth >= dept.Grade_10th_Percentage__c &&
      this.twelfth >= dept.Grade_12th_Percentage__c;
  }
  this.isEligible = eligible;
  this.resultShown = true;
  this.resultMessage = eligible
    ? 'Congratulations! You are eligible for ' + dept.Name + '. Click Apply Now to
      proceed.'
    : 'Sorry, you do not meet the eligibility criteria for ' + dept.Name +

```

```

        '. Required: 10th >= ' + dept.Grade_10th_Percentage__c + '%' +
        ', 12th >= ' + dept.Grade_12th_Percentage__c + '%' +
        (dept.Programme_Type__c === 'PG' ? ', UG >= ' + dept.UG_Percentage__c + '%' : ")
+ ':';
    }
    get resultClass() {
        return this.isEligible ? 'result-box eligible' : 'result-box not-eligible';
    }
    handleApply() {
        this[NavigationMixin.Navigate]({
            type: 'comm__namedPage',
            attributes: { name: 'Enquiry_Form__c' },
            state: { deptId: this.selectedDeptId }
        });
    }
}

```

eligibilityChecker.html

```

<template>
<div class="checker-container">
    <h2 class="title">Check Your Eligibility</h2>
    <lightning-combobox
        label="Select Department"
        placeholder="-- Choose a Department --"
        value={selectedDeptId}
        options={deptOptions}
        onchange={handleDeptChange}>
    </lightning-combobox>
    <lightning-input type="number" label="10th Percentage"
        value={tenth} onchange={handleTenth} min="0" max="100" step="0.01">
    </lightning-input>
    <lightning-input type="number" label="12th Percentage"

```

```

        value={twelfth} onChange={handleTwelfth} min="0" max="100" step="0.01">
    </lightning-input>
    <template if:true={isPG}>
        <lightning-input type="number" label="UG Percentage"
            value={ugPercent} onChange={handleUG} min="0" max="100" step="0.01">
        </lightning-input>
    </template>
    <lightning-button label="Check Eligibility" variant="brand"
        onclick={checkEligibility} class="check-btn">
    </lightning-button>
    <template if:true={resultShown}>
        <div class={resultClass}>
            <p class="result-text">{resultMessage}</p>
            <template if:true={isEligible}>
                <lightning-button label="Apply Now" variant="success"
                    onclick={handleApply}>
                </lightning-button>
            </template>
        </div>
    </template>
</div>
</template>

```

APPLICATION FORM

enquiryForm.js

```

import { LightningElement, wire, track, api } from 'lwc';
import getDepartments from '@salesforce/apex/AdmissionController.getDepartments';
import submitEnquiry from '@salesforce/apex/AdmissionController.submitEnquiry';

export default class EnquiryForm extends LightningElement {

    @api title = 'Admission Enquiry Form';

```

```

@api submitButtonLabel = 'Submit Enquiry';
// Exposed boolean property required by .js-meta.xml
// Must default to false per LWC guidelines
@api showUGField = false;

@track deptId = "";
@track studentName = "";
@track email = "";
@track mobile = "";
@track dob = "";
@track aadhaar = "";

@track tenth;
@track twelfth;
@track ugPercent;

@track isSubmitted = false;
@track enquiryNumber = "";
@track isEligible = false;
@track errorMessage = "";
@track isLoading = false;

deptOptions = [];
allDepts = [];

@wire(getDepartments)
wiredDepts({ data }) {
  if (data) {

    this.allDepts = data;

    this.deptOptions = data.map(d => ({
      label: d.Name + ' (' + d.Programme_Type__c + ')',
      value: d.Id
    }));

  }
}

// Show UG field when either administrator forces it via property OR selected department is
PG
get isPG() {
  if (this.showUGField) {
    return true;
  }
  if (!this.deptId || !this.allDepts || this.allDepts.length === 0) {
    return false;
  }
}

```



```

    }
    const selected = this.allDepts.find(d => d.Id === this.deptId);
    return selected ? (selected.Programme_Type__c === 'PG') : false;
  }

  handleDept(e) { this.deptId = e.detail.value; }
  handleName(e) { this.studentName = e.detail.value; }
  handleEmail(e) { this.email = e.detail.value; }
  handleMobile(e) { this.mobile = e.detail.value; }
  handleDob(e) { this.dob = e.detail.value; }

  handleAadhaar(e) {
    this.aadhaar = e.target.value;
    console.log('AADHAAR VALUE', this.aadhaar);
  }

  handleTenth(e) { this.tenth = parseFloat(e.target.value); }
  handleTwelfth(e) { this.twelfth = parseFloat(e.target.value); }
  handleUG(e) { this.ugPercent = parseFloat(e.target.value); }

  async handleSubmit() {

    this.errorMessage = "";

    if (!this.deptId || !this.studentName || !this.email || !this.aadhaar) {
      this.errorMessage = 'Please fill all required fields';
      return;
    }

    this.isLoading = true;

    try {

      const result = await submitEnquiry({

        deptId: this.deptId,
        studentName: this.studentName,
        email: this.email,
        mobile: this.mobile,
        dob: this.dob,
        aadharNumber: this.aadhaar,
        tenth: this.tenth,
        twelfth: this.twelfth,
        ugPercent: this.ugPercent

      });

```

B.SCREENSHOTS

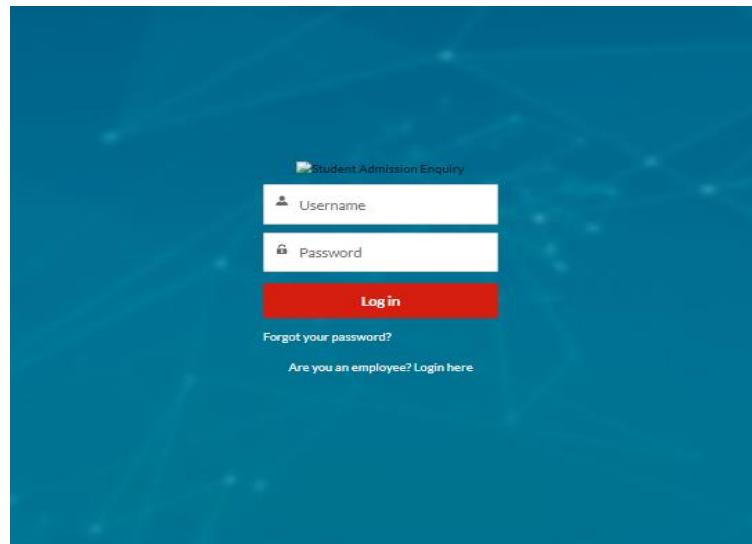


Figure B.1 Administrator Login Page

Figure B.1 , this is the login page for the administrator. They can enter the login credentials and view student details.

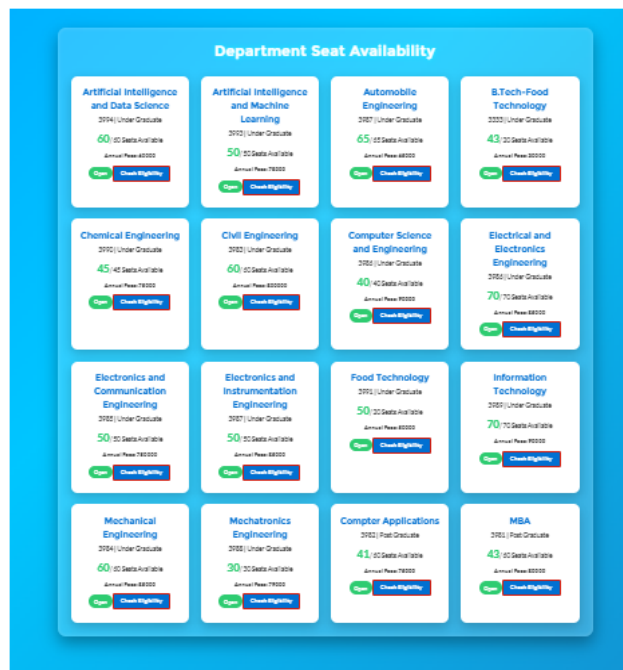
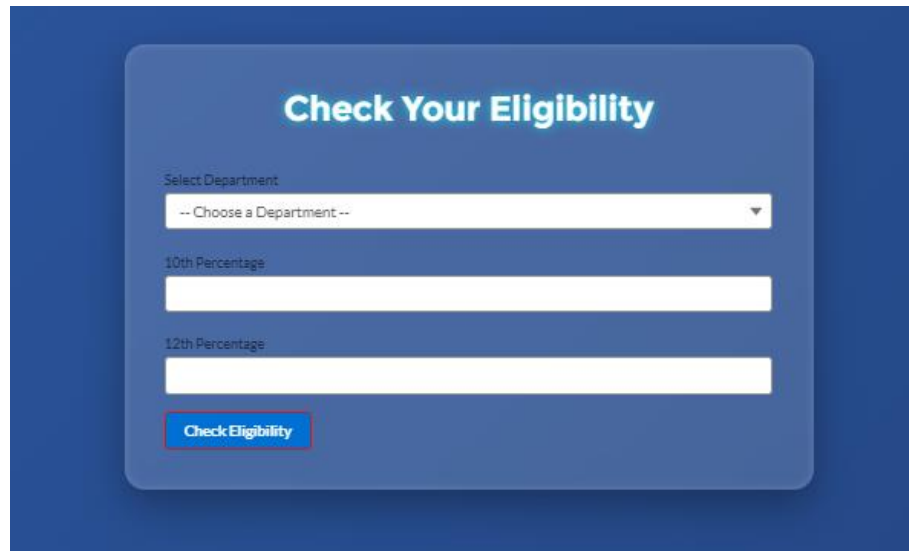


Figure B.2 Seat Availability Page

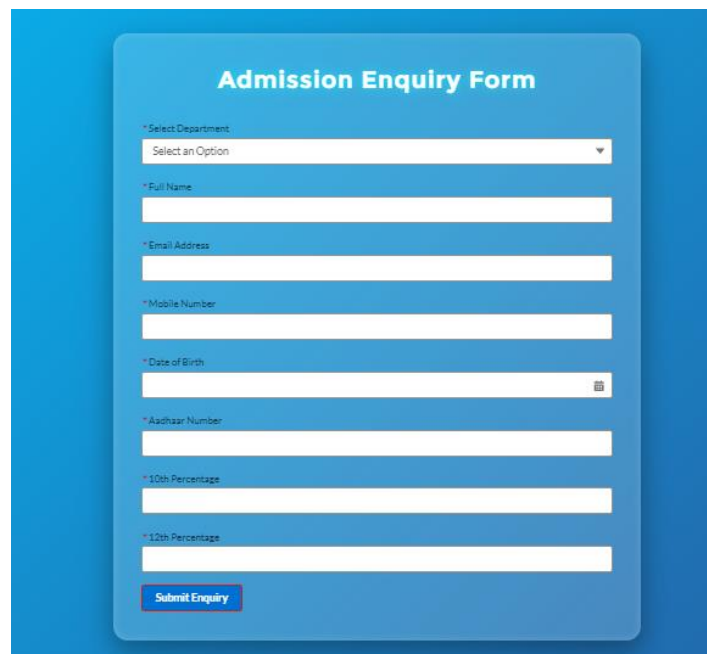
Figure B.2, the check availability page allows students to view the list of available departments along with the number of seats currently available for admission.



The image shows a web form titled "Check Your Eligibility" on a dark blue background. The form is contained within a lighter blue rounded rectangle. It features a dropdown menu for "Select Department" with the placeholder text "-- Choose a Department --". Below this are two text input fields labeled "10th Percentage" and "12th Percentage". At the bottom of the form is a blue button with the text "Check Eligibility".

Figure B.3 Check Eligibility Page

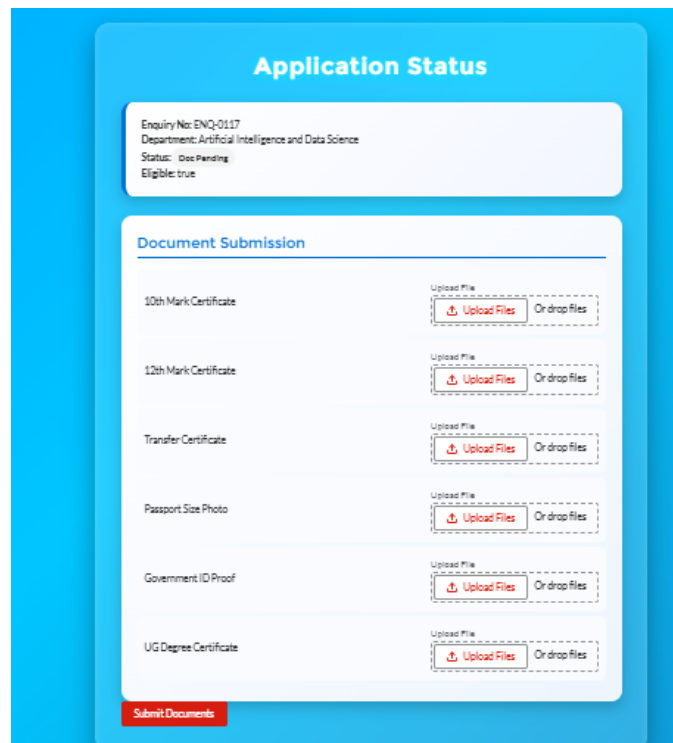
Figure B.3, the eligibility checker page allows students to verify whether they meet the minimum academic requirements for a selected department. By entering their academic percentages, the system instantly evaluates and displays whether the student is eligible to apply for that programme.



The image shows a web form titled "Admission Enquiry Form" on a blue background. The form is contained within a white rounded rectangle with a blue border. It features several input fields, each with a red asterisk indicating it is required: "Select Department" (a dropdown menu with "Select an Option" as the placeholder), "Full Name", "Email Address", "Mobile Number", "Date of Birth" (with a calendar icon), "Aadhaar Number", "10th Percentage", and "12th Percentage". At the bottom of the form is a blue button with the text "Submit Enquiry".

Figure B.4 Application Form Page

Figure B.4, the application form page allows students to submit their admission enquiry by entering personal details, academic marks, and selecting their preferred department. After submission, the system generates a unique enquiry number and displays the eligibility status of the applicant



Application Status

Enquiry No: ENQ-0117
 Department: Artificial Intelligence and Data Science
 Status: On Pending
 Eligible: true

Document Submission

Document Type	Upload File	Or drop files
10th Mark Certificate	Upload Files	
12th Mark Certificate	Upload Files	
Transfer Certificate	Upload Files	
Passport Size Photo	Upload Files	
Government ID Proof	Upload Files	
UG Degree Certificate	Upload Files	

Submit Documents

Figure B.5 Application Tracker Page

Figure B.5, the application tracker page allows students to monitor the progress of their admission application after submitting the enquiry. It displays the current status of the application, including document submission, fee payment, and final seat confirmation.



Figure B.6 Admission Dashboard Page

Figure B.6, the admission dashboard page allows administrators to monitor the overall admission process through visual reports and statistics. It displays important information such confirmed seats, department-wise seat availability.

Figure B.7 Document Verification Page

Figure B.7, the document verification form allows administrators to review and verify the documents submitted by students during the admission process. It helps ensure that all required documents are valid and correctly recorded before proceeding to the next stage.

Dear Jayamirthula,

Congratulations! 🎉

Your seat has been successfully confirmed in the department for the academic year .

Please find the important details below:

- **Department:**
- **Academic Year:**
- **Reporting Date:**
- **Reference Number:** ENQ-0001

Please report to the college campus on the reporting date with all original documents.

Contact Details:
Admission Office
Kongu Engineering College
Perundurai, Tamil Nadu
Phone: 9876543210
Email: kce@gmail.com

We look forward to welcoming you to our institution.

Regards,
Admission Office

Figure B.8 Seat Confirmation Mail

Figure B.8, the seat confirmation email is sent to students after successful fee verification to confirm their admission. It provides details such as department, reporting date, and reference number along with instructions to report to the college.

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SUSTAINABLE DEVELOPMENT GOALS (SDGs)

SDG 4 – Quality Education

The Student Admission and Enquiry Management System supports SDG 4 by improving access to quality education through an efficient and transparent admission process. By automating enquiry handling and follow-ups, the system ensures that students receive timely information and equal opportunities for enrollment, reducing the chances of missing admissions due to delays or miscommunication.

SDG 9 – Industry, Innovation, and Infrastructure

This project aligns with SDG 9 by adopting modern cloud-based Salesforce CRM technology to digitally transform the admission process. The system strengthens institutional infrastructure by introducing automation, scalability, and secure data management, promoting innovation in educational administration.

SDG 16 – Peace, Justice, and Strong Institutions

The system contributes to SDG 16 by promoting transparency, accountability, and efficiency in admission management. Centralized data storage, role-based access, and accurate reporting reduce errors and bias, helping institutions build trustworthy and well-governed admission processes.